Source: hierarchy A

Goal: hierarchy $A \cup C$

A_1 a_1 a_2

A_2 a_1 a_2 a_3 a_4

A_3 a_1 a_2 a_3 a_4 a_5 a_6 a_7 a_8

$\tilde{S}_1$? ?

$\tilde{S}_2$? ? ? ?

$\tilde{S}_3$? ? ? ? ? ? ? ?

A_1 a_1 a_2 → $\tilde{S}_1$ a_1 a_2

case 1: directly fill from A_1

A_2 a_1 a_2 a_3 a_4 — $\tilde{S}_2$? ? ? ?

A_3 a_1 a_2 a_3 a_4 a_5 a_6 a_7 a_8 — $\tilde{S}_3$? ? ? ? ? ? ? ?

cascade

A_1 a_1 a_2 — $\tilde{S}_1$ a_1 a_2

case 2: cascade fill from A_i

A_2 ~~a_1 a_2 a_3 a_4~~ — $\tilde{S}_2$ a_1 a_2 a_6 a_7

A_3 ~~a_1 a_2 a_3 a_4 a_5~~ a_6 a_7 a_8 — $\tilde{S}_3$? ? ? ? ? ? ? ?

cascade

A_1 a_1 a_2 — $\tilde{S}_1$ a_1 a_2

case 3: cascade fill from pool $A \cup C$

A_2 a_1 a_2 a_3 a_4 — $\tilde{S}_2$ a_1 a_2 a_6 a_7

A_3 ~~a_1 a_2 a_3 a_4 a_5 a_6 a_7~~ a_8 — $\tilde{S}_3$ a_1 a_2 a_6 a_7 a_8 p_1 p_2 p_3

p_1 p_2 p_3 p_x p_x p_x p_x p_x

FPS over the pool $A \cup C$